

Day: 10

* SMTP — Simple Mail Transfer Protocol

- SMTP is used to send emails from client to server and between mail servers.
- Port 25 — SMTP (server to server)
- Port 587 — SMTP (Client to server, modern)
- Port 465 — SMTP over SSL

* SMTP Conversation

Client → EHLO mail.example.com

Server → 250 Hello

Client → Mail FROM:

<sender@example.com>

Server → 250 OK

client → RCPT TO:

<receiver@example.com>

Server → 250 OK

Client → DATA

Server → 354 Start Input

Client → Subject: Hello
From: sender@example.com
[email body]

Server → 250 OK Message accepted

Client → QUIT

Server → 221 Bye

* SMTP in DFIR

- Phishing emails travel over SMTP — you can see the full email in plaintext if unencrypted.
- MAIL FROM can be spoofed — attacker puts any address here.
- RCPT TO shows who the email was actually delivered to
- Email headers reveal the true origin IP of the sender.
- Wireshark filter: smtp

* FTP — File Transfer Protocol

- FTP is used to transfer files between client and server.
- Port 21 — FTP Control (commands)

- Port 20 — FTP data (actual file transfer)
- Completely unencrypted — credentials and files visible in plaintext!

* FTP Commands

- USER → send username
- PASS → Send password — visible in plaintext
- LIST → List files
- RETR → Download a file
- STOR → Upload a file
- QUIT → Disconnect

* FTP Conversation

Client → USER abhishek

Server → 331 Password required

Client → PASS mypassword123 ← **Visible in WIRESHARK!**

Server → 230 Login Successful

Client → LIST

Server → [directory listing]

Client → RETR secret.txt

Server → [file contents sent]

* FTP in DFIR

- FTP credentials are fully visible in Wireshark — use filter: ftp
- File contents are also visible — use filter: ftp-data
- Attackers use FTP to infiltrate stolen data
- Look for STOR commands — files being uploaded OUT
- Look for RETR commands — files being downloaded
- Wireshark filter: ftp

* SSH — Secure Shell

- SSH is used for secure remote login and encrypted file transfer.
- Port 22 — SSH
- Everything is encrypted — you cannot see commands or data in Wireshark.
- Replaces Telnet (port 23) which was unencrypted

* SSH in Wireshark:

- You can see the connection was made (IP + Port 22)
- You can see packet sizes and timing
- You cannot see the actual commands or data
- Filter: ssh or tcp.port == 22

* What SSH protects:

- Telnet (port 23) → Everything visible in plaintext
- SSH (port 22) → Everything encrypted

* SSH in DFIR

- SSH connections to unusual IPs = possible backdoor.
- SSH on non-standard ports (not 22) = suspicious.
- Large SSH data transfers = possible exfiltration over encrypted channel.
- Attackers use SSH tunneling to bypass firewalls.
- Even though encrypted, connection metadata is valuable.

* Protocol Comparison Table

Protocol	Port	Encryption	Purpose	DFIR Risk
SMTP	25/587	None (usually)	Send email	Phishing, Spoofing
FTP	21/20	None	File Transfer	Credentials visible
SSH	22	Full	Secure Remote	Tunneling, Backdoor
Telnet	23	None	Remote Login	Everything visible
IMAP	143	None	Receive Email	Email Theft
POP3	110	None	Receive Email	Email Theft

* Diagrams:

- SMTP vs FTP vs SSH

SMTP — Port 25/587 (unencrypted)	FTP — Port 21/20 (unencrypted)
SSH — Port 22 (fully encrypted)	



- Secured

- SMTP Conversation

EHLO mail.example.com
MAIL FROM: sender
RCPT TO: receiver
DATA
[email body]
QUIT

- FTP Conversation : Password Visible!

USER abhishek
PASS mypassword
230 Login OK
LIST
RETR secret.txt
QUIT

- SSH Connection

TCP connect port 22
Key exchange
Encrypted Tunnel
Commands hidden
Data hidden

- DFIR — what to look for

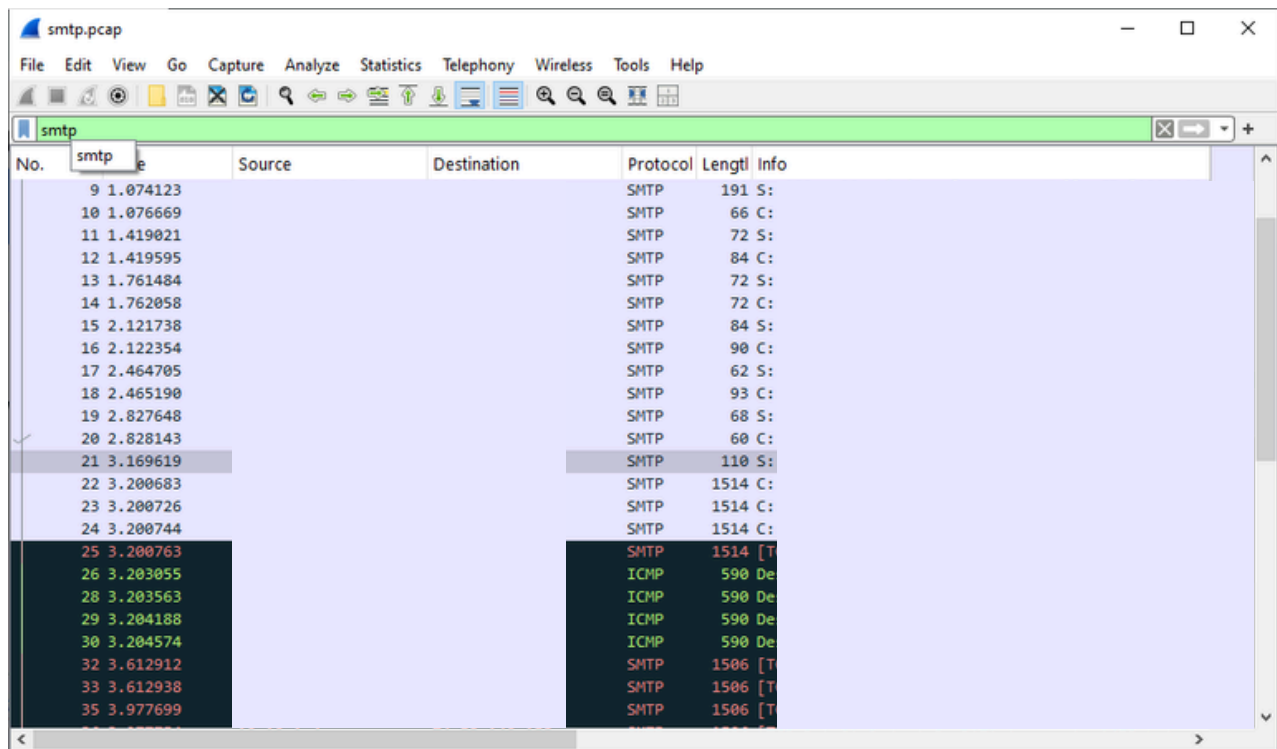
SMTP Phishing spoofed FROM address	FTP Credentials USER + PASS in Plaintext	SSH Tunneling Unusual IPs, Large Data
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Go to <https://wiki.wireshark.org/SampleCaptures>

Download the smtp.pcap file

Open it in Wireshark

Apply filter: **smtp**



The image shows the Wireshark interface with the 'smtp.pcap' file loaded. The filter bar at the top is set to 'smtp'. The packet list pane displays a table of captured packets, with columns for No., Time, Source, Destination, Protocol, Length, and Info. The table is filtered to show only SMTP packets. Packet 21 is highlighted in blue, and packet 25 is highlighted in black.

No.	Time	Source	Destination	Protocol	Length	Info
9	1.074123			SMTP	191	S:
10	1.076669			SMTP	66	C:
11	1.419021			SMTP	72	S:
12	1.419595			SMTP	84	C:
13	1.761484			SMTP	72	S:
14	1.762058			SMTP	72	C:
15	2.121738			SMTP	84	S:
16	2.122354			SMTP	90	C:
17	2.464705			SMTP	62	S:
18	2.465190			SMTP	93	C:
19	2.827648			SMTP	68	S:
20	2.828143			SMTP	60	C:
21	3.169619			SMTP	110	S:
22	3.200683			SMTP	1514	C:
23	3.200726			SMTP	1514	C:
24	3.200744			SMTP	1514	C:
25	3.200763			SMTP	1514	[T
26	3.203055			ICMP	590	De
28	3.203563			ICMP	590	De
29	3.204188			ICMP	590	De
30	3.204574			ICMP	590	De
32	3.612912			SMTP	1506	[T
33	3.612938			SMTP	1506	[T
35	3.977699			SMTP	1506	[T



The image shows the 'Follow TCP Stream' window in Wireshark, displaying the SMTP conversation between the client and the server. The window title is 'Wireshark - Follow TCP Stream (tcp.stream eq 0) - smtp.pcap'. The stream content is as follows:

```
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

EHLO GP

AUTH LOGIN

235 Authentication succeeded

MAIL FROM:

250 OK

RCPT TO:

250 Accepted

DATA

354 Enter message, ending with "." on a line by itself

From:
To: <
Subjec
Date:
Messag
MIME-V
Conten
X-Mail
Conten
x-c-r-h
BBAHWA
x-c-r-p

This i
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Conten
```